File Processing Systems and Database Systems

What is DATA?

- facts and statistics stored or free flowing over a network, generally it's raw and unprocessed
- becomes information when it is processed.

File System vs Database System

File System

Database System

File System

A file system is a software application that organizes and maintains files on a storage device. It manages the storage and retrieval of data.

DBMS

A database management system, or DBMS, is a software application that allows you to access, create, and manage databases.

File System	DBMS
It does not provide data recovery services.	There is a backup recovery option for data.
There is no crash recovery mechanism in the file system.	A crash recovery method is provided by DBMS.

File System

It’s difficult to keep a file system safe.

Centralization process is difficult.

DBMS

A good protection method is provided by DBMS.

Centralization is simple to achieve.

File System	DBMS
It has a higher level of data inconsistency.	It has a low level of data inconsistency.
Data redundancy is higher.	Redundancy is minimized to a great extent.

File System	DBMS
There is no efficient query processing.	Data can be retrieved based on the requirements and can be easily filtered from the database.
File systems allow only one user to access data at a time.	Data can be accessed simultaneously by multiple users.

File System

NTFS (New Technology File System), FAT (File Allocation Table), EXT (Extended File System), etc.

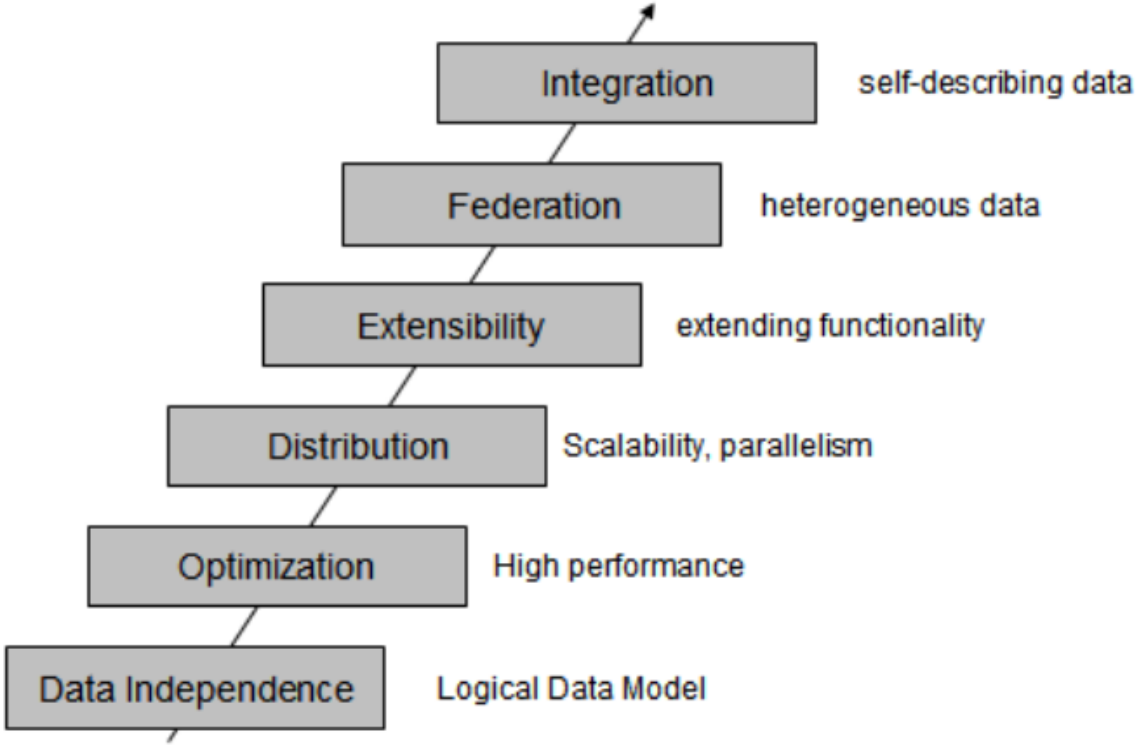
DBMS

MySQL, Oracle, SQL Server, PostgreSQL, Microsoft SQL Server, Microsoft Access, etc.

Database

- A database is a repository of data, designed to support efficient data storage, retrieval and maintenance.
  - Relational database – tabular form
  - Hierarchical database – structure form
  - Network database – graphs representing relationships between objects.

Evolution of DBMS

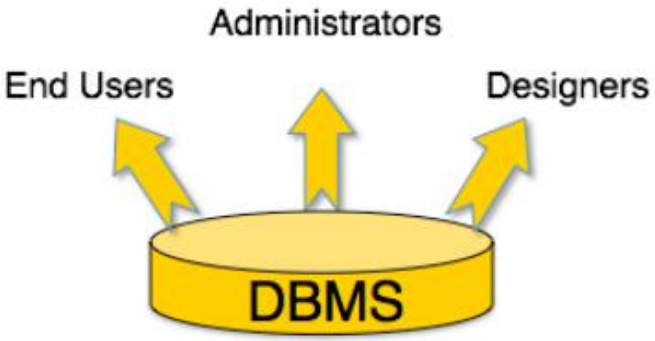


DBMS Characteristics

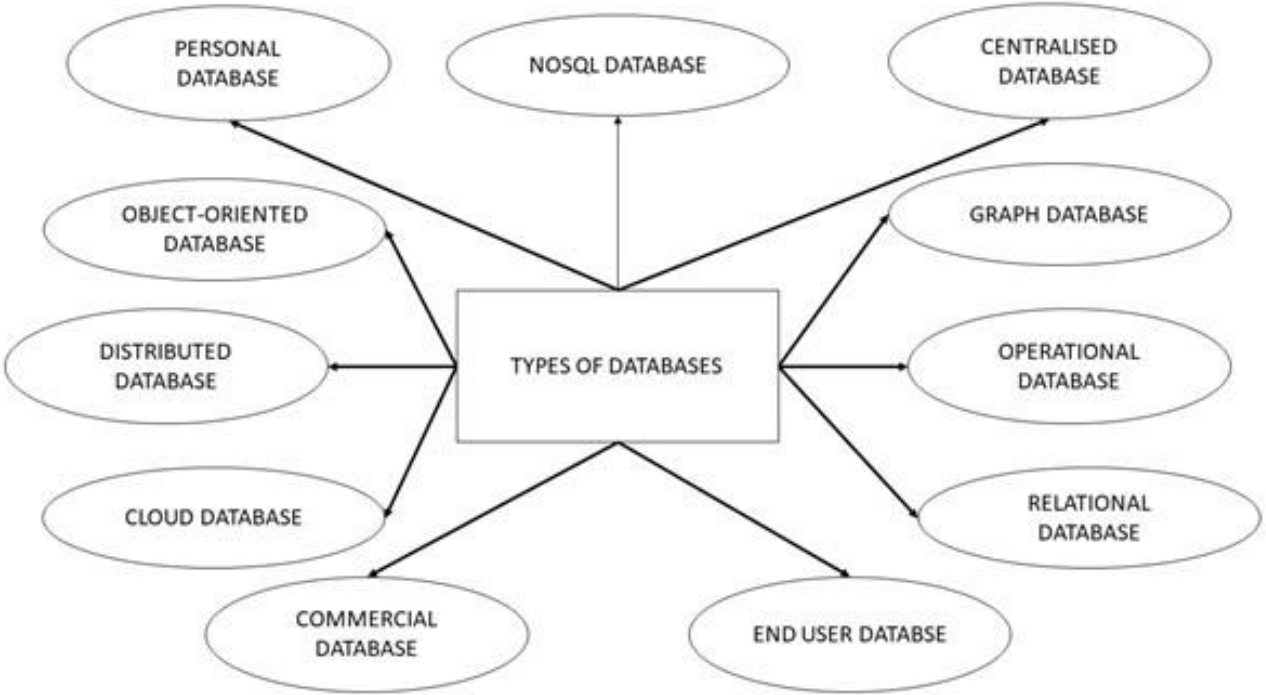
- Real-world entity
- Relation-based tables
- Isolation of data and application
- Less redundancy
- Consistency
- Query language
- ACID properties - Atomicity, Consistency, Isolation and Durability
- Multiuser and concurrent access
- Multiple views
- Security

DBMS Users

- Administrators
  - Create access profile for users, apply limitations to maintain isolation and force security.
- Designers
  - Keep a close watch on what date should be kept and in what format
- End Users
  - Make use of the DBMS



Types of Database



Database Applications

- Banking: transactions
- Airlines: reservations, schedules
- Universities: registration, grades
- Sales: customers, products, purchases
- Online retailers: order tracking, customized recommendations
- Manufacturing: production, inventory, orders, supply chain
- Human resources: employee records, salaries, tax deductions